Quantification: 536-63x_Hyp_RD_z08c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	19
interior lacuna count	19
frame-edge lacuna count	0

Interior lacunae only: mean, median and sample SD (n = 19)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2669.84	2736	1499.86	px^2
major_axis_length_px	lacuna shape	99.0415	102.242	40.5286	px
minor_axis_length_px	lacuna shape	39.1288	39.2919	11.2588	px
aspect_ratio	lacuna shape	2.5275	2.5123	0.884	unitless
eccentricity	lacuna shape	0.8845	0.9174	0.0757	unitless
solidity	lacuna shape	0.7772	0.8492	0.1359	unitless
roots_count	headline	11.6316	10	6.1029	count
ring_length_r30_px	headline	338.105	332	167.303	px
ring_length_r60_px	reported	669.474	647	333.703	px
owned_length_px	ownership-dependent	990.54	954.264	756.365	px
edge_count	ownership-dependent	47.6316	45	35.0305	count
mean_edge_length_px	ownership-dependent	20.4964	20.4868	4.5101	px
perimeter_px	normalised	303.303	274.735	131.113	px
ring_area_r30_px2	normalised	9484.95	10082	3265.38	px^2
ring_area_r60_px2	normalised	21315.1	22623	7346.24	px^2
in_frame_fraction_r30	normalised	0.9864	1	0.0448	unitless
in_frame_fraction_r60	normalised	0.9728	1	0.0801	unitless
ring_density_r30	normalised	0.0346251	0.0338051	0.00738597	px^-1
ring_density_r60	normalised	0.0304341	0.030342	0.00720463	px^-1
roots_per_100px_perimeter	normalised	3.8293	3.8422	0.9639	per 100 px
ring_attached_length_r30_px	network v2	312.789	308	163.131	px
ring_attached_length_r60_px	network v2	542.737	566	304.476	px
ring_length_w_r30_px	network v2	382.329	377.463	188.641	px
ring_length_w_r60_px	network v2	761.905	733.441	375.252	px
sholl_crossings_r10	network v2	8.2632	9	3.0884	count
sholl_crossings_r20	network v2	8.9474	10	4.564	count
sholl_crossings_r30	network v2	8.7368	8	4.1344	count
loop_count	loops, report only	2.6316	1	3.715	count
loop_edge_share	loops, report only	0.1482	0.1471	0.1209	share
ring_width_mean_r30_px	width	7.3248	7.3169	0.7563	px

Per field
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measure	value	units
analysed_area_px2	945450	px^2
total_skeleton_length_px	26151	px
canalicular_length_density_per_px	0.0276598	px^-1
junction_count	509	count
junction_density_per_px2	0.00053837	px^-2
lacuna_count	19	count

lacunae_per_px2	2.01e-05	px^-2
skeleton_component_count	350	count
mean_component_length_px	74.7171	px
median_component_length_px	23.5	px
field_length_density_w_per_px	0.0314271	px^-1
field_density_without_flagged_per_px	0.026229	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	297	px
unattached_component_count	16	count
canalicular_length_density_attached_per_px	0.0276598	px^-1
field_length_density_with_unattached_per_px	0.027974	px^-1
excluded_area_px2	52399	px^2
analysed_field_area_px2	996177	px^2
loop_count	58	count
loop_edge_share	0.1588	share
edge_count	1373	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.7532	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	26138	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2181	148.407	31.7971	4.6673	0.9768	0.5491	-0.2241	127.792	674.16	False
2	3003	146.448	39.2919	3.7272	0.9633	0.5868	-0.1342	219.3	634.066	False
3	4713	113.091	66.6058	1.6979	0.8082	0.6869	0.0149	166.949	811.934	False
4	2838	120.116	56.4992	2.126	0.8825	0.5006	-0.0922	183.402	482.588	False
5	593	40.0782	21.7723	1.8408	0.8396	0.7583	0.4372	249.45	834.877	False
6	5444	169.703	43.0843	3.9389	0.9672	0.8492	-0.1663	315.438	989.012	False
7	5032	145.049	50.087	2.8959	0.9385	0.6975	-0.3567	389.803	259.63	False
8	1325	56.519	31.2238	1.8101	0.8335	0.9138	-0.4181	431.483	840.339	False
9	1869	63.382	43.4291	1.4594	0.7284	0.783	-0.4231	489.012	823.251	False
10	4023	130.997	44.5634	2.9396	0.9404	0.8529	-0.2947	478.273	373.626	False
11	2063	91.1753	31.0575	2.9357	0.9402	0.8892	-0.3885	511.078	505.802	False
12	2736	102.242	36.178	2.8261	0.9353	0.8653	-0.4465	632.235	931.07	False
13	2738	96.5558	37.4594	2.5776	0.9217	0.8784	-0.3793	679.917	399.446	False
14	3288	105.667	42.059	2.5123	0.9174	0.8761	-0.3815	724.683	177.336	False
15	797	46.6745	23.1748	2.014	0.868	0.8935	-0.2375	760.548	828.605	False
16	584	33.791	23.1054	1.4625	0.7297	0.9125	-0.7852	793.788	813.11	False
17	1183	68.5981	39.5227	1.7357	0.8173	0.5796	-0.4587	845.742	872.086	False
18	4334	128.945	44.2583	2.9135	0.9393	0.8796	-0.1396	903.815	374.562	False
19	1983	74.3475	38.2775	1.9423	0.8573	0.8154	-0.0273	970.974	775.836	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	23	400	663	757.997	48	15.7916	7.9397
2	14	379	809	954.264	57	16.7415	6.7363
3	22	486	824	1102.66	45	24.5036	7.3511
4	19	723	1433	1324.53	81	16.3522	6.5938
5	6	202	455	766.826	32	23.9633	7.3169
6	11	339	594	1189.9	53	22.451	7.3398
7	22	665	1296	3298.37	161	20.4868	6.9583
8	6	119	250	103.841	8	12.9801	6.9542
9	8	191	416	367.191	15	24.4794	6.9373
10	14	462	941	1193.56	68	17.5524	8.5326
11	10	330	647	2084.07	79	26.3807	7.9422
12	6	212	490	509.558	29	17.571	8.2498
13	14	332	705	1145.63	45	25.4585	7.7637
14	10	342	853	1599.81	60	26.6635	7.5793
15	4	138	264	294.25	19	15.4868	5.5078
16	4	120	171	195.38	8	24.4225	7.1011
17	11	289	435	428.161	30	14.272	7.1548
18	10	424	931	993.666	41	24.2358	8.6364
19	7	271	543	510.588	26	19.638	6.5754

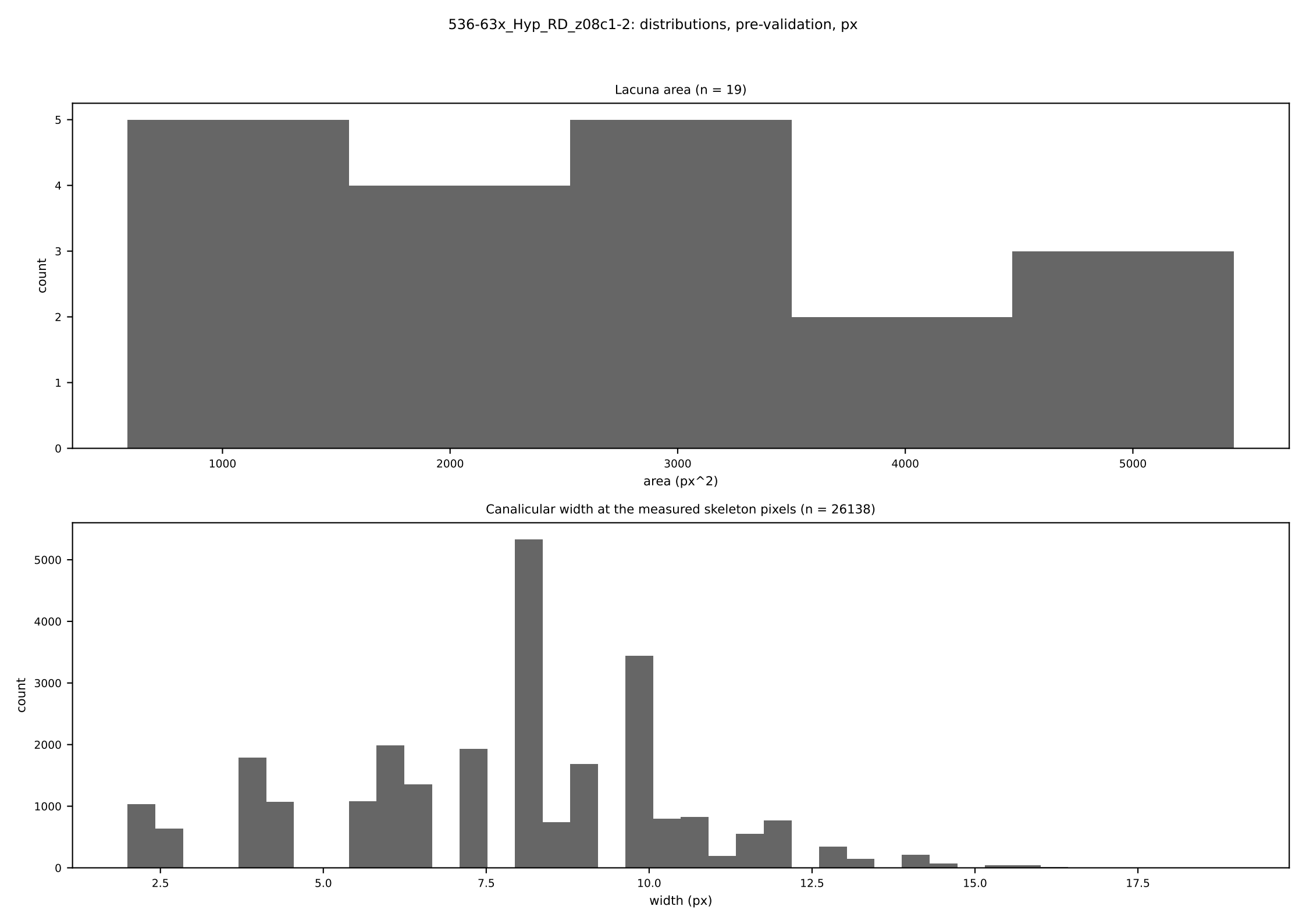
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	415.641	10082	20733	1	1	0.0396747	0.031978	5.5336
2	495.973	10286	21637	1	1	0.0368462	0.0373897	2.8227
3	442.446	12800	26586	1	1	0.0379688	0.0309938	4.9724
4	514.931	14099	31464	1	1	0.0512802	0.0455441	3.6898
5	128.539	4684	10910	1	1	0.0431255	0.0417049	4.6678
6	423.747	11743	23622	0.8152	0.7003	0.0288683	0.025146	2.5959
7	487.973	14467	32198	1	1	0.0459667	0.040251	4.5084
8	156.16	5263	12719	1	1	0.0226107	0.0196556	3.8422
9	228.202	6707	15935	1	1	0.0284777	0.0261061	3.5057
10	348.433	12410	27080	1	1	0.037228	0.0347489	4.018
11	227.38	9187	22623	1	1	0.0359203	0.0285992	4.3979
12	274.735	10452	25593	1	1	0.0202832	0.0191459	2.1839
13	250.693	9821	24082	1	1	0.0338051	0.029275	5.5845
14	298.534	10681	26849	1	1	0.0320195	0.0317703	3.3497
15	131.776	4471	9578	1	1	0.0308656	0.0275632	3.0355
16	102.847	3623	7408	1	1	0.0331217	0.0230832	3.8893
17	259.557	9006	18739	1	1	0.0320897	0.0232136	4.238
18	340.877	12243	29334	1	0.982	0.034632	0.0317379	2.9336
19	234.309	8189	17896	0.9258	0.8017	0.0330932	0.030342	2.9875

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	351	447	450.505	755.583	9	12	10
2	368	654	434.333	925.566	10	8	8
3	458	760	549.032	940.465	10	11	12
4	694	1052	813.127	1613.81	14	11	16
5	202	435	228.924	526.002	6	3	5
6	339	566	384.948	682.097	10	10	7
7	597	1169	751.855	1472.74	13	19	16
8	105	105	133.225	279.551	6	0	2
9	150	205	210.853	471.789	3	6	5

10	461	903	523.232	1073.89	10	13	13
11	308	587	377.463	733.441	12	8	10
12	149	242	242.693	562.328	6	10	7
13	332	637	382.363	821.637	8	10	10
14	281	583	365.723	944.382	9	11	11
15	138	244	155.225	303.764	4	4	2
16	96	110	136.711	200.965	4	4	5
17	272	399	334.635	500.073	7	7	7
18	408	822	478.747	1043.47	10	16	13
19	234	392	310.664	624.642	6	7	7


536-63x_Hyp_RD_z08c1-2: distributions, pre-validation, px
Lacuna area (n = 19)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 26138)
count
width (px)